In vitro mass balance for PFAS: a primer from first principles

What this is. Everything in `MBM_Qin_vs_Armitage.md` rests on one equation and about ten approximations. This document derives the equation, names every approximation, shows where each one breaks, and then reads the five papers against it — two models and three datasets.

Read it with `scripts/mbm_published_equations.py` open. Every number quoted here comes from that file, and every code block below runs as written.

Part I — The one equation

1. What a partition coefficient actually is

Put a chemical in contact with two phases, water and something else. At equilibrium the chemical potential is equal in both:

```
mu_A = mu_B          mu_i = mu_i° + RT ln(gamma_i x_i)
```

Rearranging, the ratio of mole fractions is

```
x_A / x_B = (gamma_B / gamma_A) · exp( -(mu_A° - mu_B°) / RT )
```

The exponential is a constant at fixed temperature. So the ratio is constant **if and only if the activity coefficients *gamma* are constant**. Convert mole fractions to concentrations and you have the partition coefficient

```
K_AB = C_A / C_B                  (constant)
```

> ***Approximation 1 (dilute / linear regime).*** *gamma is constant only when the chemical is a trace. Add enough of it and the sorbent changes — sites fill, the phase swells, the chemical starts interacting with itself. Then K is no longer a constant and becomes an apparent ratio that falls as you add more. This is the single most important approximation in the whole subject. Qin's model is built around relaxing it; Armitage's is not.*

Notation. `K` is used for a true constant; `D` for a *distribution ratio*, which is what you measure when the chemical exists in more than one form (here: neutral and anionic) or when the sorbent is a mixture (a cell, serum). `D` is what appears in every balance below.

2. The well as a set of phases

A well in a 96- or 24-well plate contains, from the model's point of view:

phase	symbol	what it is
medium water	V_w	the aqueous phase, where "free" concentration lives
serum albumin	V_alb	from the FBS in the medium
serum lipid	V_slip	from the FBS
cells	V_cell	protein + lipid + cell water
plastic	A_plastic	the well wall and base, a surface not a volume

Each has a distribution ratio to water, `D_i = C_i / C_w`.

3. Deriving the balance

Total moles are conserved and distributed over the phases:

```
n_total = C_w·V_w + C_alb·V_alb + C_slip·V_slip + C_cell·V_cell + C_pl·A_pl
```

Substitute `C_i = D_i · C_w` in every term — this is the step that uses Approximation 1, five times:

```
n_total = C_w · ( V_w + D_alb·V_alb + D_slip·V_slip + D_cell·V_cell + D_pl·A_pl )
```

Define the **total sorptive capacity**

```
S ≡ V_w + Σ_i D_i · V_i                                  [litres of water-equivalent]
```

and the balance is a one-liner:

```
C_w = n_total / S                                        (★)
```

Everything else follows. The concentration in the cells is `C_cell = D_cell·C_w`, and the **fraction of the nominal dose that ends up inside cells** is

```
x_cell = C_cell·V_cell / n_total = D_cell·V_cell / S     (★★)
```

That is the whole model. Both papers are (★★) with different recipes for the `D_i`.

4. The approximation ledger

Everything that had to be true to get from the well to (★):

#	approximation	breaks when	does it bite here?
A1	linear sorption, *gamma* constant	sorbent saturates	**yes** — Qin measures K_d = 0.317 µM for PFOA against doses of 6.25–100 µM
A2	equilibrium reached	slow uptake	no — Fragki's time course reaches steady state in ~1 h, exposure is 24 h
A3	no metabolism or degradation	metabolised compounds	no for PFCA/PFSA; **yes** for fluorotelomer alcohols
A4	no active transport	transporter-expressing cells	unknown — HepG2 express few PFAS transporters, but this is assumed, not shown
A5	phases are independent and additive	one sorbent alters another	assumed
A6	volumes additive and constant	—	fine
A7	no losses (evaporation, tubing, pipettes)	volatiles	fine for ionised PFAS
A8	the aqueous phase is pure water	co-solvents present	**3% ethanol / 0.5% / 0.1% DMSO** — see §6
A9	the cell is a well-mixed bag of phases	organelles, lysosomal trapping, pH gradients	assumed; Armitage's published model has no lysosome term
A10	one `D_cell` describes all cells equally	different cell lines	**yes** — three assays, two cell lines, one composition used for all

A1, A8 and A10 are the ones that have actually cost us something in this project. A4 is the one nobody can check.

5. Two axes, and a trap worth naming

There are two ways to report the answer:

```
C_cell  = D_cell · n_total / S                    [concentration]
x_cell  = D_cell · V_cell / S                     [fraction of dose]
```

Read those as written and `x_cell` is the one carrying `V_cell`. That is correct — and it is the opposite of what a careless reading of "which axis needs the cell volume" suggests, because the question has two sides.

On the model side. `x_cell ∝ V_cell` directly. `C_cell` contains `V_cell` only inside `S`, where the cell term is 0.1–18% of the total, so predicted `C_cell` is nearly volume-free.

On the measurement side, converting a published "nmol per mg protein":

```
measured x_cell = (nmol/mg) · (mg protein in the well) / (nmol added)       no volume
measured C_cell = (nmol/mg) · (mg protein per cell) / (litres per cell)     needs volume
```

The volume dependence does not disappear — it changes sides. And the quantity actually tested, predicted ÷ measured, is *identical* on the two axes. Rosenmai PFOA at 100 µM, holding protein per cell fixed at 694 pg and changing only the volume:

ng/cell	meas x_cell	pred x_cell	ratio	meas C	pred C	ratio
10.0	0.330%	0.660%	**2.00×**	182 µM	364 µM	**2.00×**
3.04	0.330%	0.201%	**0.61×**	599 µM	366 µM	**0.61×**

> ***Choosing the fraction axis buys no protection against the cell-volume uncertainty.*** *Both ratios scale as (volume per cell)/(protein per cell). That is exactly why §14 of the memo can move Armitage's bias from 3.87× to 1.84× by swapping 10 ng/cell for 3.04 pL/cell — it is a real lever, not an artefact of the axis.*

What the fraction axis does buy is that the *measurements* become mutually comparable with no volume assumption at all. Rosenmai, Amstutz and Fragki can be put on one plot and read against each other directly (fig. 20A). On the concentration axis they could not be, because each would need a cell volume nobody measured.

A cancellation that does work — sometimes. Cell *number* per well is also uncertain (HepG2 divide between seeding and exposure; Fragki never state the count for differentiated HepaRG). Whether it cancels depends on what the paper reported:

	reports	measured x_cell	ratio over a 4× change in N
Fragki	nmol/mg protein	built from N, so ∝ N	**8.69× -> 7.89×**, cancels
Rosenmai	uptake % directly	a fixed published number	**1.00× -> 3.97×**, does not

So the memo's §8 remark — that the ratio is insensitive to cell number — holds for Fragki, which is where it is made, and **not** for Rosenmai, whose Table 1 is already a percentage. For Rosenmai the assumed 5.43×10^5 (three wells at the *seeded* 1.8×10^5) propagates straight into the comparison.

Part II — Three results you can derive in two lines each

These are the levers that explain most of what we found. Each is a direct consequence of (★★); none needs a model.

6. Why a co-solvent almost cancels

A co-solvent makes water a better solvent, so **every** water->phase partition coefficient falls by roughly the same factor `k`. Put `D_i -> D_i/k` into (★★) and work it through. Split the capacity into its water part and the rest, `S = V_w + Σ` with `Σ ≡ Σ_i D_i V_i`, and write `w ≡ V_w/S`, so `Σ = S(1-w)`:

```
x'_cell = (D_cell/k)·V_cell / ( V_w + Σ/k )

x'_cell     (D_cell/k)·V_cell        S              1          S
-------  =  ----------------- × ------------  =   ---  ×  ----------
x_cell       V_w + Σ/k           D_cell·V_cell      k      V_w + Σ/k
```

Substitute `V_w = wS` and `Σ = S(1-w)`; every `S` cancels:

```
x'_cell        1              1                  1                   1
-------  =    ---  ×  ----------------  =  -------------  =  ---------------
x_cell         k       w + (1-w)/k           k·w + 1 - w       1 + (k-1)·w
```

which is

```
x'_cell / x_cell = 1 / ( 1 + (k-1)·w )              (★★★)
```

Read it off: the numerator falls by `k`, and so does the denominator — but only on the `(1-w)` of it that carries a `D`. The mismatch is exactly water's share.

Three sanity checks, all immediate: `k = 1` gives 1 (no co-solvent, no effect); `w = 0` gives 1 (nothing dissolved in water, total cancellation); `w = 1` gives `1/k` (no sorbents at all, the full effect). And it is **exact**, not a linearisation — checked against the model to machine precision for all five compounds.

- `w -> 0` (long chains, albumin dominates): no effect at all.
- `w -> 1` (short chains, water dominates): the full `1/k`.

Check it against the real run. Rosenmai, Armitage, 3% ethanol ⇒ k ~ 1.35:

	w	(★★★) predicts	model gives
PFBA	0.80	1/(1+0.35·0.80) = **0.781**	1.11/1.42 = **0.782**
PFNA	0.005	1/(1+0.35·0.005) = **0.998**	9.31/9.33 = **0.998**

Exact to five decimals.

> ***The approximation inside the result.*** *(★★★) assumes one* `k` *for every phase. Real cosolvency powers sigma differ by sorbent, so strictly* `D_cell -> D_cell/k1` *and* `D_alb -> D_alb/k2`*. In the albumin-dominated limit the ratio becomes ~* `k2/k1`*, so what survives is the difference in sigma, not sigma itself. A full unit of sigma difference at 3% ethanol (sigma_cell = 5 vs sigma_alb = 4) is worth 10^(0.03×1) = 1.07, i.e. 7%. Still small against 5×, but it is not identically zero the way the single-*`k` *version is.*

An error that multiplies every sorptive term cancels on this axis. Only an error in the *relative* size of two sinks survives.

7. Why albumin's share sets the sensitivity

Let `a ≡ D_alb·V_alb / S` be albumin's share of the capacity. Multiply the albumin coefficient by `f` (an error):

```
S' = S·(1 - a + f·a)        so      x'_cell / x_cell = 1 / (1 - a + f·a)   (★4)
```

Two things follow, and it is worth being careful about which is which.

(i) How much a wrong albumin *coefficient* costs you. For a 10×-too-low one (`f = 0.1`), (★4) is `1/(1 - 0.9a)`. In Fragki's well:

	a at 0% FBS	a at 2%	a at 10%	cost at 2%	cost at 10%
PFBA	0.000	0.046	0.195	1.04×	1.21×
PFHxA	0.000	0.687	0.917	2.62×	5.71×
PFOA	0.000	0.884	**0.974**	4.89×	**8.09×**
PFNA	0.000	0.791	0.948	3.47×	6.80×

At 0% serum the albumin parameter is irrelevant and the number is exactly 1.00 by construction. **A model validated on serum-free data says nothing about its behaviour at 10% FBS** — and the same model is 4–8× more sensitive to its albumin parameter in a 10% FBS well than in a 2% one.

(ii) How much a wrong albumin *volume* costs you, which is the Fragki case. Coding 10% FBS where the paper says 2% leaves the coefficient alone and multiplies `V_alb` by 5 — equivalent to `f = 0.2` going from the true well to the coded one. (★4) with the true `a` then predicts the whole error:

	a (true, 2% FBS)	(★4) with f = 0.2	actual Armitage
PFBA	0.046	1.19×	1.18×
PFHxA	0.687	3.75×	3.75×
PFOA	0.884	**4.53×**	**4.54×**
PFNA	0.791	4.14×	4.17×

Three lines of algebra reproduce a 4.5× modelling error to two decimals. That is what it means to say the balance is an albumin calculation with a cell attached.

> *Note* `a` *is evaluated in the reference system — the one you are coming from. Use the coded system's* `a` *instead and you get the inverse factor, not the same number, because (★4) is not symmetric in* `f`.

8. Why the residual's chain-length slope is a *ratio*

Once albumin dominates (`a` = 0.94–0.97 from PFOA up), `S ~ D_alb·V_alb`, so

```
x_cell ~ D_cell·V_cell / (D_alb·V_alb)
log x_cell ~ log(D_cell/D_alb) + log(V_cell/V_alb)
```

The second term has no chain-length dependence. Differentiate with respect to chain length `n` and subtract the same for the measurement:

```
d(log x_pred/x_meas)/dn  =  d log(D_cell/D_alb)_model/dn - d log(D_cell/D_alb)_true/dn   (★5)
```

The residual's slope is the error in the slope of the ratio — not in either coefficient alone. A model can have both coefficients wrong and no residual trend (if the errors track), or both nearly right and a strong trend (if they don't). This is exactly what separates the two models; §13 below.

Part III — The two models

Both are (★★). They differ entirely in how they produce the `D_i`.

9. Armitage 2021 (IV-MBM EQP v2.0)

Strategy: predict every `D` from one descriptor, log K_ow.

9a. Ionisation

PFAS at pH 7.4 are essentially fully anionic. Armitage splits each `D` into a neutral and an ionic part:

```
D_ow = [ F_n·10^logKow + F_i·10^(logKow - d_OW) ] / 10^(-k_salt·c_salt)
```

with `F_n` the neutral fraction, `F_i = 1 - F_n`, and `d_OW` the penalty the anion pays. The denominator is a salting-out correction, `k_salt = 0.04·logKow + 0.114` at `c_salt = 0.15 M`.

For PFAS `F_n = 0`, so every term reduces to the anionic branch. The penalties in SI Table S1:

phase	penalty `d`
octanol	3.1
membrane lipid	1.0
serum albumin	**0.0**
plastic	3.1

> ***Approximation A11.*** `d_SaW_acid = 0` *says the anion binds albumin exactly as well as the neutral form would. That is a modelling choice, not a measurement, and it is a large part of why albumin ends up holding 94–97% of the well in this model.*

9b. The spLFERs

"Sorbent-phase-specific linear free energy relationships" — one regression per phase, from SI S1:

```
log K_mw = 1.01·logKow + 0.12                    membrane lipid–water
log K_pl = 0.97·logKow - 6.94                    plastic–water
log K_sa = 1.08·logKow - 0.70     (logKow < 4.5)   albumin, lower branch
log K_sa = 0.37·logKow + 2.56     (logKow >= 4.5)   albumin, upper branch
```

> ***Approximation A12.*** *log K_ow is a sufficient descriptor. For a perfluorinated surfactant with an anionic head this is the weakest link in the model: PFAS are neither simply hydrophobic nor simply lipophilic, and their K_ow is itself hard to measure.* ***Approximation A13.*** *The albumin spLFER is piecewise, flattening to a slope of 0.37 above log K_ow 4.5 — calibrated on neutral organics and pharmaceuticals, where a plateau is physically sensible because the binding cavity has finite depth. Whether PFAS–albumin binding plateaus there is an open question, and §13 shows the consequences of assuming it does.*

9c. The cell as a sum of phases

```
D_cell/w = f_sl·D_ow + f_ml·D_mw + 0.035·f_prot·D_ow + f_water·1
```

Four terms: storage lipid behaves as octanol; membrane lipid as a phospholipid membrane; **3.5% of the protein behaves as octanol and the rest is inert**; the cell's own water behaves as water.

The AREc2 composition (SI Tables S3–S6), used for all three assays:

```
f_sl = 0.00046    f_ml = 0.00414    f_prot = 0.0514    f_water = 0.944
```

> ***Approximation A14, the big one.*** *"3.5% of bulk protein behaves as octanol" is a convention imported from the neutral-organics literature. Measured against Qin 2024's protein–water ratios it is* ***457–7,820× too low****.* ***Provenance caveat, added on audit.*** *The 0.035 in this implementation comes from EPA's* `httk`*, and the code flags it as not found in Armitage's SI S1. So "Armitage's convention" — as §13–14 and §16 of the memo call it — is an attribution this project has not verified at source. The consequence is a property of the implementation either way: with a protein term this small the cell is lipid, and a lipid cell has too steep a chain-length slope. But whose convention it is remains open, and it matters, because if Armitage's own model treats cell protein differently then the §16 verdict lands on* `httk` *rather than on the paper.* ***Resolving this needs SI S1 §2, which is not in hand.*** *The consequence: the protein term contributes 0.1–0.2% of Armitage's cell, so the cell is 79% water at PFBA and* ***99.7% membrane lipid at PFNA****. The convention has defined protein out of the cell.*

10. Qin 2023

Strategy: measure the `D`s, and let albumin saturate.

10a. The Langmuir, derived

Take a protein with `n` identical, independent binding sites. For one site:

```
S + L <=> SL          K_a = [SL] / ([S]·[L])          K_d ≡ 1/K_a
```

Sites are conserved, `[S]_tot = [S] + [SL]`, so the occupancy is

```
theta = [SL]/[S]_tot = K_a[L] / (1 + K_a[L]) = [L] / (K_d + [L])
```

Now put it on a per-volume-of-albumin basis. The concentration of **sites inside pure albumin** is just the molar concentration of albumin times n:

```
C_max = n · rho_alb / M_alb = n · 1364 g/L ÷ 66,463 g/mol = n × 20,523 µM
```

So bound ligand per litre of albumin is `C_b = C_max·theta`, and the distribution ratio is

```
D_specific = C_b / C_w = C_max / (K_d + C_w)
```

Add a linear, non-saturating term for everything that is not a defined site:

```
D_BSA/w = C_max / (K_d + C_w) + D_non-specific              Qin Eq. 16   (★6)
```

Check the derivation against Qin's own table. He reports C_max both per litre of BSA and per mole of BSA; the ratio must be 20,523 µM:

	C_max [µmol/L_BSA]	C_max [mol/mol]	ratio
PFBA	9,220	0.450	20,489
PFOA	25,400	1.24	20,484

0.2% from the derived value. The derivation is right, and n = 0.45 for PFBA and 1.24 for PFOA — **less than one site for PFBA**.

10a-bis. How badly Approximation 1 breaks here

A1 does not break *in* Qin's model. **Qin's model is what you build because A1 broke in the experiment** — he measured the isotherm, found it curved, and wrote (★6) to carry the curvature. Armitage keeps A1; Qin abandons it. Here is how much was at stake.

Albumin has about one site. From the C_max identity above, n = 0.45 (PFBA), 1.24 (PFOA), 0.92 (PFHxS), 0.39 (PFOS). So albumin's high-affinity capacity for PFAS is bounded at roughly **one mole per mole of albumin**.

Now count the albumin actually in the well. Armitage's serum albumin is 24 g/L, so the medium holds `24 × f_FBS` g/L:

	FBS	albumin	doses	nominal PFAS : albumin
Rosenmai	10%	2.40 g/L = **36.1 µM**	10–100 µM	0.28 – 2.77 mol/mol
Amstutz	10%	2.40 g/L = **36.1 µM**	100 µM	2.77 mol/mol
Fragki	2%	0.48 g/L = **7.2 µM**	6.25–100 µM	0.87 – **13.85 mol/mol**

Against `n ~ 1`, a ratio of 2.8 means roughly three times more PFAS than there are sites. A ratio of 13.9 means fourteen times. For scale, Fischer 2024 worked at <= 0.004 mol/mol and estimated the ratio in human serum at <= 0.0005 — so **these assays run 5,500–27,700× above the physiological ligand:protein ratio.** A1 is not marginally violated here; it is nowhere near true.

What that does to the "constant". Rearranging (★6), the specific term is just its own trace limit times a saturation penalty:

```
D_specific = C_max/(K_d + C_w) = D_trace,spec · 1 / (1 + C_w/K_d)
```

That factor `1/(1 + C_w/K_d)` **is** the activity-coefficient correction of Approximation 1, written for a site-binding sorbent. *gamma* constant means "the sorbent the Nth molecule meets is the same one the first molecule met". Fill the sites and it is not.

Between the trace limit and full saturation, `D_BSA/w` falls to the non-specific floor:

	K_d (µM)	D at trace	D at the floor	collapse
PFBA	18.3	546	43	12.8×

	K_d (µM)	D at trace	D at the floor	collapse
PFOA	0.317	85,100	5,010	17.0×
PFHxS	0.0744	256,000	2,950	**86.6×**
PFOS	0.00624	1,350,000	69,200	19.6×

And inside the actual wells, solving the balance self-consistently:

study	cmpd	dose	C_free	C_free/K_d	occupancy	D / D_trace
Rosenmai	PFOA	10 µM	0.081 µM	0.26	20%	0.81×
Rosenmai	PFOA	30 µM	0.419 µM	1.3	57%	0.46×
Rosenmai	PFOA	100 µM	5.30 µM	16.7	**94%**	**0.11×**
Fragki	PFOA	6.25 µM	0.367 µM	1.2	54%	0.50×
Fragki	PFOA	100 µM	22.9 µM	72.4	**99%**	**0.07×**

A parameter that is supposed to be a constant moves 7-fold inside a single dose series. That is the whole justification for Qin's extra machinery.

Four consequences that run through the entire comparison:

1. **The dose exponent (§4 of the memo).** Armitage is linear by construction, pinned at b = 1.00. Measured b runs 0.85–1.42 and rises with chain length. Only a saturating model can produce that, and Qin does.
2. **Qin's error swings with dose** where Armitage's does not — his saturation is fitted to BSA isotherms, and the well's saturation need not match it.
3. **A saturating sink amplifies an error in its own size.** Cut the albumin 5× and Armitage's prediction moves by exactly the denominator arithmetic, 4.54× at every dose. Qin's moves by up to 14×, because less albumin -> more free -> higher occupancy -> lower D_BSA -> less capacity -> more free:

Fragki PFOA, 10% -> 2% FBS	occupancy	D_BSA ratio	Qin x_cell	Armitage x_cell
6.25 µM	13% -> 54%	0.56×	**7.89×**	4.54×
25 µM	48% -> 93%	0.23×	**14.17×**	4.54×
100 µM	94% -> 99%	0.63×	4.56×	4.54×

The feedback is largest **mid-range**, where the 10% well is still linear and the 2% well has already tipped into saturation. At the top dose both are saturated and it vanishes.

4. **A literature binding constant is meaningless without its *nu*.** An apparent K_a measured at *nu* = 3 is not the number measured at *nu* = 0.004 — Qin 2024 later measured both regimes and found specific binding exceeds non-specific by 3.0–5.0×, with the changeover around *nu* ~ 0.01 to *nu* ~ 1. This is why the binding database in `ALBUMIN_BINDING.md` records the ligand:protein ratio for every row it can.

```
import sys; sys.path.insert(0, "tk_review/scripts")
import mbm_published_equations as P, mbm_qin_vs_armitage as Q

p = Q.qin_params("PFOA", 7)
trace = p["Cmax"] / p["Kd"] + 10 ** p["logDnonspec"]
a = P.ASSAYS["Fragki"]; V = P.volumes(a)
alb = P.SI["conc_ser_alb"] * a["FBS"] / Q.BSA_MW * 1e6      # uM albumin
for d in a["doses"]:
    r = P.qin_run("PFOA", d, V)
    theta = r["Cw"] / (p["Kd"] + r["Cw"])
    print(f"{d:6g} uM   nominal PFAS:alb {d/alb:5.2f}   C_free {r['Cw']:7.3f} uM"
          f"   occupancy {theta:6.1%}   D/D_trace {r['Dbsa']/trace:5.2f}x")
```

10b. The trace limit, and the bridge to the binding literature

When `C_w << K_d`, (★6) collapses to a constant:

```
D_trace = C_max/K_d + D_ns = n·(rho_alb/M_alb)·K_a + D_ns
```

Dropping the non-specific term and converting to a **mass** basis (divide by *rho*_alb):

```
log D [L_w/kg_protein] = log K_a + log n - log10(M_protein/1000)          (★7)
```

For albumin (M = 66,500) the constant is **1.823**. This is the bridge that makes the entire binding literature — reported as K_a in M^{-1} — comparable with a partition coefficient in L/kg. Without it, the two cannot even be compared.

> ***Approximation A15.*** *(★7) needs* n*, the stoichiometry, which is almost never reported with a K_a. It moves log D by up to 0.78 across the plausible range.*

10c. Qin's cell

```
D_cell/w = D_SP/w · f_protein + D_lip/w · f_lipid       — and NO water term
```

`D_SP/w` is a *measured* non-specific protein–water ratio. The cell is 92–95% protein at every chain length.

> ***Approximation A16.*** *Omitting cell water means Qin's cell cannot fall below its protein term, which is why his PFBA cell partition (log 0.38) sits above Armitage's (log 0.08) — Armitage's is almost pure cell water there.*

11. The two models, side by side

	Armitage 2021	**Qin 2023**
source of `D`	predicted from log K_ow	measured, then interpolated
albumin	linear spLFER, piecewise	**saturating Langmuir**
albumin anion penalty	0	n/a (measured on the anion)
cell	lipid + 3.5% protein-as-octanol + water	protein + lipid, no water
cell at PFBA	79% water	92% protein
cell at PFNA	**99.7% membrane lipid**	95% protein
dose dependence	linear by construction	non-linear
compounds parameterised	any, from log K_ow	PFBA, PFOA, PFHxS, PFOS only

> ***Approximation A17.*** *Qin measured four compounds. PFHxA and PFHpA are interpolated between the two carboxylate anchors (PFBA, PFOA) and* ***PFNA is extrapolated beyond both****. Only* `log D_lip/w` *has a published interpolation rule (Fig. A1,* `0.476·nCF + 0.172`*); everything else is a two-point interpolation with no basis in the paper.*

12. Where the dose actually goes — run this

```
import sys; sys.path.insert(0, "tk_review/scripts")
import mbm_published_equations as P, mbm_qin_vs_armitage as Q

a = P.ASSAYS["Rosenmai"]; V = P.volumes(a)
print(f"{'cmpd':7s}{'albumin':>10s}{'ser.lipid':>11s}{'water':>9s}"
      f"{'cells':>9s}{'plastic':>9s}{'  a (alb share)':>16s}")
for nm, _nC, nCF, _mw, ko, _kc, _s in Q.COMPOUNDS:
    lk = ko if ko is not None else Q.KOW_OPERA(nCF)
    D = P.armitage_D(lk, V["comp"])
    terms = {"albumin":   D["Dsa"] * V["Valb"],
             "ser.lipid": D["Dow"] * V["Vslip"],
             "water":     V["Vw"],
             "cells":     D["Dcw"] * V["Vcells"],
             "plastic":   D["Dpl"] * V["sarea"] * 1000.0}
    S = sum(terms.values())
    print(f"{nm:7s}" + "".join(f"{terms[k]/S:>{w}.1%}" for k, w in
          (("albumin",10),("ser.lipid",11),("water",9),("cells",9),("plastic",9)))
          + f"{terms['albumin']/S:>16.3f}")
```

Running it gives a = 0.196, 0.918, 0.977, 0.982, 0.971 for PFBA->PFNA, with water at 80.3% for PFBA and 0.5% for PFNA. **The balance is a water calculation for the shortest chain and an albumin calculation for everything else** — and the switch happens between PFBA and PFHxA, in one step of two CF_2. That abrupt switch is why PFBA behaves like a different compound in every analysis in this project, and why it has so much leverage on any regression across the series.

13. Why the two models fail differently

Apply (★5). Over nCF 5–8, in decades per CF_2:

	albumin	cell	ratio	residual, observed
Armitage	+0.386	+0.769	**+0.383**	**+0.345**
Qin	+0.547	+0.515	**-0.032**	**-0.074**
Fischer 2024, *measured* BSA	**+0.540**	—	—	—

(★5) predicts the residual slope from the ratio slope, and it does — to within the noise on a four-point fit.

Armitage is wrong from both ends and the errors stack. Its albumin rises at +0.386 where Fischer measures +0.540 (A13, the piecewise flattening); its cell rises at +0.769 where a flat ratio needs ~ +0.540 (A14, the cell is lipid). 40% of the gap is the spLFER, 60% the cell.

Qin gets both right — albumin within 0.007 of Fischer's measurement — and has essentially no chain-length residual, despite being *worse* on the level.

> ***Qin has the right chain-length physics and the wrong level. Armitage has the better level and the wrong chain-length physics.***

Part IV — The three data papers

All three measure the same quantity on the same axis: **moles found in the cells ÷ moles added to the well**. They differ in almost everything else.

14. Rosenmai et al. 2018 — *J Appl Toxicol* 38:219, doi 10.1002/jat.3515

What. 14 PFAS in HepG2; cellular concentration by LC-MS/MS alongside a PPAR*alpha* reporter assay; the point of the paper is the *relationship* between cellular concentration and receptor activation.

For us. §2.5: 12-well plates, 18×10^4 cells/well, 24 h at 10/30/100 µM, 5× ice-cold PBS wash, lysates from **three wells pooled**. Table 1 is the uptake percentage — the 15 points this project uses. Recovery 78–98%.

Gotchas.

- Stocks are in **ethanol**, and §2.4 holds it at **3% v/v** — the largest co-solvent of the three. (§6 shows why it nearly cancels.)
- The **working volume is never stated.** 1 mL/well is an assumption.
- The 3% ethanol is stated for the PPAR*alpha* assay; §2.5 does not repeat it.
- Uptake is < 1% for everything except PFDoDA, PFTeDA and FOSA.

Read: §2.1, §2.2, §2.4 (the ethanol sentence), §2.5, §3.3 + Table 1.

15. Amstutz et al. 2022 — *Toxicology* 480:153312, doi 10.1016/j.tox.2022.153312

What. 13 PFAS in HepG2; MTT viability and DCFH reactive-oxygen assays at 0–800 µM, plus intracellular PFAS at 100 µM. The point is a structure–activity ranking for cytotoxicity.

For us. §2.5: 96-well, 4×10^4 cells, 250 µL dosed, 50 µL taken at t = 0 for a mass balance, 24 h on the remaining 200 µL, 5× DPBS. Exposure is in **the culture medium — DMEM + 10% FBS**, 0.1% DMSO carrier. Explicitly "based on the

study of Rosenmai et al. (2018), with small modifications".

Gotcha. Amstutz reports an *intracellular fraction* normalised to **recovered** mass, not to nominal. Its denominator excludes whatever went to the plastic, so its numbers are not quite on the same axis as the other two.

Read: §2.2, §2.5. Skip MTT/DCFH entirely.

16. Fragki et al. 2022 — *Food Chem Toxicol* 172:113559, doi 10.1016/j.fct.2022.113559

What. A QIVIVE case study: PFOA, PFNA, PFHxS, PFOS in differentiated HepaRG, with cell-associated concentrations fed through a PBK model to oral equivalent doses. The biokinetics are a means to that end.

For us. The dose series in nmol/mg protein, 6.25–100 µM, Supplementary Table 2. §2.3: 24-well, 550 µL dosed, 50 µL at t = 0, 5× DPBS. The time course reaches steady state at **~1 h** — which is what licenses A2.

Gotchas.

- **§2.2: exposure is in "assay medium (growth medium containing 2% FBS)"**, not the 10% of the growth medium. This single sentence was worth ~4× on every Fragki prediction.
- Cells are differentiated in **1.7% DMSO for twelve days** and held there — hepatocyte-like and triglyceride-loaded, nothing like the AREc2 composition they are modelled with (A10, A14).
- The authors themselves **excluded PFNA at 100 µM** from their QIVIVE for low protein content suggesting cytotoxicity. It is the largest value in the set.

Read: §2.2 (last sentence), §2.3, §3.1, Supplementary Table 2.

17. What makes the three mutually inconsistent

	Rosenmai	Amstutz	Fragki
cells	HepG2	HepG2	**differentiated HepaRG**
exposure FBS	10%	10%	**2%**
co-solvent	**3% ethanol**	0.1% DMSO	0.5% DMSO
plate	12-well ×3 pooled	96-well	24-well
volume	**not stated**	250 -> 200 µL	550 -> 500 µL
normalised to	nominal	**recovered**	nominal

And the measurements themselves disagree: at 100 µM PFHxA, Rosenmai reads 0.14% and Amstutz 0.039% — **3.6× apart on the same cell line and the same axis**. No model error explains that one.

Part V — The same equation, in serum

Fischer et al. 2024 (*ES&T* 58:1055, doi 10.1021/acs.est.3c07415) applies (★) to serum instead of a well. Serum = water + albumin + globulin:

```
D_serum/w = VF_w + D_HSA/w·c_alb + D_glob/w·c_glob            [L_w / L_serum]
f_unbound = VF_w / D_serum/w
```

Watch the units. `D_HSA/w` is in L_water per **kg** of albumin, `c_alb` in kg per litre of serum, so the product is L_water per litre of serum. The protein **density cancels**: writing it as Fischer does, `D_vol·VF` with `VF = c/rho` and `D_vol = D_mass·rho`, the *rho* drops out. It survives only in `VF_w`.

This is the one place in the whole subject where the balance is *validated*: 26 PFAS have D_HSA/w, D_glob/w **and** D_serum/w measured separately, and predicting the third from the first two with no fitted parameter gives a geometric-mean bias of 1.00× and a typical error of 1.55× (`ALBUMIN_BINDING.md` §13a). The structure works. It is the in vitro *well* that does not.

Check your understanding

1. In (★★), `x_cell = D_cell·V_cell/S`. If you doubled the number of cells in the well, would the measured fraction of dose in cells double? Under what condition?

2. The co-solvent result (★★★) says the effect scales with `w = V_w/S`. Rosenmai's PFBA has `w = 0.80`. What does that tell you about which compound you should use to *detect* a co-solvent problem experimentally?

3. Qin reports `n` = 0.45 binding sites for PFBA. A site count below one is physically odd. Give two readings of it — one that is a real property of the system, one that is an artefact of the fit.

4. Armitage's albumin anion penalty is `d_SaW_acid = 0` while its octanol penalty is 3.1. What would happen to albumin's share `a` if the albumin penalty were also 3.1, and what would that do to the model's predicted uptake?

5. Both models were given the AREc2 cell composition for Fragki's differentiated HepaRG. HepaRG carry visible triglyceride droplets. Which of Armitage's four cell terms would that change most, and would it make the over-prediction better or worse?

6. (★5) says the residual's chain-length slope is the error in the slope of `D_cell/D_alb`. Suppose someone measured a new albumin series and found a slope of +0.70 per CF_2 instead of Fischer's +0.540. Would that make Armitage's chain-length problem better or worse — and what would it do to Qin's?

Answers

1. Only if the cells are a negligible share of `S`. `x_cell = D_cell·V_cell/ (V_w + … + D_cell·V_cell)`. When the cell term is small the denominator barely moves and `x_cell ∝ V_cell` — doubling cells doubles uptake. When cells hold a large share, the denominator grows too and the fraction saturates toward 1. In these assays the cell share is 0.1–18%, so the proportionality roughly holds — which is also why the *ratio* predicted/measured is insensitive to the cell count, and why §8 of the memo says the comparison survives not knowing it.

2. The opposite of what you would guess. The compound where water dominates is where a co-solvent shows up — so you test it with **PFBA**, not PFNA. For a long chain the effect cancels to 0.2%. Generally: to detect an error that multiplies all the `D`s, look where the one term without a `D` is large.

3. (a) Real: `n` is an *apparent* mean site count over a heterogeneous population of sites with different affinities; a weak, partly-occupied class of sites fitted as one homogeneous class returns `n < 1`. (b) Artefact: `C_max` and `K_d` are strongly correlated in a Langmuir fit, so if the measured concentration range does not reach saturation, the pair is only determined along a ridge and `n` is not separately identifiable. For PFBA `K_d` = 18.3 µM against doses up to 100 µM, so saturation *is* sampled — which favours reading (a).

4. `D_sa` would fall by 10^3.1 ~ 1,260×, so `a` would collapse from ~0.95 to near zero and the cell and plastic would carry the well. Free concentration would rise by up to the same factor and the model would over-predict uptake catastrophically. This is the sense in which `d_SaW_acid = 0` is doing enormous work: it is the assumption that makes albumin the dominant sink.

5. Storage lipid, `f_sl`, which is 0.00046 in AREc2 — essentially zero. Triglyceride droplets are storage lipid, so the real `f_sl` could be orders above it. That raises `D_cell`, raises `x_cell`, and makes the over-prediction **worse**. Like every other correction in this project, the direction is unhelpful — which is itself the finding.

6. Armitage **better**, Qin **worse**. Armitage's ratio slope is `cell - albumin` = 0.769 - 0.386 = +0.383; a steeper true albumin slope means Armitage's +0.386 is further below truth, so more of the residual is explained and a corrected albumin would flatten more of it. Qin's +0.547 would now be 0.15 *below* a true +0.70, opening a ratio-slope error it does not currently have. The point of the exercise: the ratio is what matters, so "better albumin data" helps or hurts depending entirely on which model you are holding.

Where to go next

file	what
`report/MBM_Qin_vs_Armitage.md`	the full comparison, §15 the medium correction, §16 the chain-length residual
`report/ALBUMIN_BINDING.md` §12–13a	the K_d ↔ partition bridge (★7) and the serum balance that works
`scripts/mbm_published_equations.py`	both models, from the published equations
`scripts/mbm_medium_composition.py`	§6 and §7 of this primer, run against the data
`scripts/mbm_chain_length_residual.py`	§8 and §13 of this primer